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7 March 1983
SUPERSEDING
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25 May 1979

MILITARY SPECIFICATION

WATER QUALITY ANALYSIS SET: PREVENTIVE MEDICINE

This specification is approved for use by the Mobility Equipment Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a water quality analysis set for use by preventive medicine personnel.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-535	- Plastic Sheet (Sheeting); Plastic Strip; Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
NN-P-71	- Pallets, Material Handling, Wood, Stringer Construction, 2 Way and 4 Way.
QQ-S-781	- Strapping, Steel, and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Mobility Equipment Research and Development Command, ATTN: DRDME-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6630

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MMM-A-1617	- Adhesive, Rubber Base, General Purpose.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-C-4150	- Cases, Transit and Storage, Waterproof and Water-Vaporproof.
MIL-H-9884	- Honeycomb Material, Cushioning, Paper.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52929	- Enamel, Alkyd, Camouflage, Flash Dry.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-147	- Palletized and Containerized Unit Loads, 40-inch X 48-inch Pallets, Skids, Runners or Pallet Type Base.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-882	- System Safety Program for Systems and Subsystems and Equipment; Requirements for.
MIL-STD-1188	- Commercial Packaging of Supplies - and Equipm. .
MIL-STD-1472	- Human Engineering Design Criteria for Military Systems, Equipment and Facilities.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

TA13222E7900 - Water Quality Analysis Set, Preventive Medicine.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which

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are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI Z129.1-1976 - Precautionary Labeling of Hazardous Industrial Chemicals.

(Applications for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The water quality analysis set (hereinafter called "set") shall consist of test kits, equipment, and reagent chemicals necessary to conduct water analysis in the field. The set shall be in accordance with Top Assembly TA13222E7900 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the set. Any data (shop drawings, layouts, flow sheets, processing procedures etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available upon request for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The contractor shall furnish one or more sets as specified (see 6.2) for examination and testing within the time frame specified (see 6.2) to prove that his production methods and choice of design detail and components will produce sets that comply with the requirements of this specification. Examination and tests shall be as specified in Section 4; and, unless otherwise specified herein, all examinations and tests shall be conducted by the contractor subject to surveillance and approval by the Government (see 6.3). When specified (see 6.2), the Government will conduct any or all of the preproduction examination and tests, as specified (see 6.2).

3.3 Initial production. When specified (see 6.2), the contractor shall furnish to the Government one or more sets for inspection as specified in 4.4.

3.4 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.6).

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3.4.1 Glass. Glass or other easily broken material shall not be used except for optical reasons, or when specifically requested or approved by the Government.

3.4.2 Degradable materials. Materials that are deformed by or deteriorate after prolonged exposure to water or reagent chemicals supplied in the set shall not be used.

3.4.3 Chemical reagents. Chemical reagents may be of either the liquid or dry form. Liquid reagents used in drop-count method of analysis shall be packaged in bottles with droppers or in self-contained dropper bottles. The reagents shall be formulated so that not more than 30 drops of liquid or 30 additions of dry reagent are required to determine any value within the range specified for each test.

3.5 Test Kits.

3.5.1 Acidity test kit. The test kit (see drawing 13222E7890) shall provide the equipment and chemical reagents necessary to determine the total acidity of water in the range of 7 to 500 milligrams per liter (mg/l) (as $CaCO_3$). A drop count titration method using a standard sodium hydroxide solution and Brom Cresol Green/Methyl Red Indicator Powder will be used. The sodium hydroxide titrating agent shall be of such strength that one drop of reagent shall be equal to 17 mg/l (milligrams per liter) when using a 5.83 ml sample.

3.5.1.1 Performance. The test kit shall be capable of performing not less than 100 tests at 20 mg/l.

3.5.2 High-range chloride test kit. The test kit (see drawing 13222E7892) shall provide the equipment and chemical reagents necessary to determine the chloride content of water in the range of 1000 to 20,000 mg/l. A silver nitrate solution drop count method shall be used with a chloride indicator. The test reagent shall be of such concentration that one drop shall be equal to 606 mg/l chloride (CL) or 1000 mg/l sodium chloride (NACL), with a 5.83 sample.

3.5.2.1 Performance. The test kit shall be capable of performing not less than 100 tests at 3000 mg/l of chloride.

3.5.3 Iron test kit. The iron test kit (see drawing 13222E7896) shall consist of a comparator, two color disks, and other equipment and chemical reagents necessary to determine the iron content in water. One color disk shall have the range of 0 to 1.0 mg/l and the other from 0 to 10.0 mg/l. A color comparator disc shall be used for ferric iron determination with a phenanthrolene indicator powder for color development. The strength of the indicator powder shall be such as to develop sufficient color for an accurate iron determination of 0 to 1.0 mg/l iron when using a 25 ml sample.

3.5.3.1 Performance. The test kit shall be capable of performing not less than 100 tests at 0.3 mg/l Iron.

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3.5.4 Dissolved oxygen test kit. The test kit (see drawing 13222E7895) shall provide the equipment and chemical reagents necessary to determine the dissolved oxygen content of water in the range of 0.5 to 15 mg/l. A manganese sulfate/alkali iodide-azide reagent shall be used together with a standard sodium thiosulfate titrant for determination of dissolved oxygen. The thiosulfate reagent shall be of such strength that 1 ml of the reagent shall be equivalent to 4 mg/l dissolved oxygen with a 65 ml sample.

3.5.4.1 Performance. The test kit shall be capable of performing not less than 50 tests at 2 mg/l.

3.5.5 Zinc test kit. The test kit (see drawing 13222E7915) shall provide the equipment and chemical reagents necessary to determine the zinc content of water in the range of 1 to 20 mg/l. A zinc test reagent powder together with a zinc conditioning reagent powder shall be used for determination of zinc in water. The zinc reagent powder shall be of such strength that 0.5 gms of the reagent shall be sufficient to develop an indicator color to be compared to a standard calibrated color block with a range of 0-10 mg/l zinc using a 2.0 ml water sample.

3.5.5.1 Performance. The test kit shall be capable of performing not less than 100 tests at 20 mg/l.

3.5.6 Multipurpose test kit. The multipurpose test kit (see drawing 13222E7912) shall consist of a battery-operated direct-reading spectrophotometer in a carrying case plus all equipment and chemical reagents required to perform the following tests in the ranges specified:

(a) pH - 4.0 to 10.0 units.
(b) Fluoride - 0.1 to 2.0 mg/l.
(c) Iron (Fe^{++}) - 0.1 to 2.0 mg/l.
(d) Iron (Fe^{+++}) - 0.1 to 2.0 mg/l.
(e) Ammonia nitrogen - 0.1 to 2.0 mg/l.
(f) Nitrate nitrogen - 1.0 to 30.0 mg/l.
(g) Sulfate - 5.0 to 150 mg/l.
(h) Turbidity - 20.0 to 500.0 mg/l.

The spectrophotometer shall utilize a circular variable interference filter with a wavelength range of 400 to 700 nanometers (nm) and shall have an internal battery supply. Batteries shall be included with the kit. The method for determining ferric iron (Fe^{+++}) may be accomplished by determining total iron and subtracting the amount of ferrous iron (Fe^{++}). The equipment and chemical reagents shall be packaged separately in the set case.

3.5.6.1 Performance. The test kit shall include enough chemical reagents to perform not less than 100 tests at each of the following concentrations:

(a) pH - 10.0 units.
(b) Fluoride - 1/5 mg/l.

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(c) Iron (Fe^{++}) - 0.3 mg/l.
(d) Iron (Fe^{+++}) - 0.3 mg/l.
(e) Ammonia nitrogen - 0.5 mg/l.
(f) Nitrate nitrogen - 10.0 mg/l.
(g) Sulfate - 50 mg/l.

3.6 Carrying case.

3.6.1 Set case. All items comprising a set shall be packaged in a case conforming to MIL-C-4150, type I, style 1, class A, variety NHC except those changes as specified in the drawings or as specified herein. Cushioning material as specified in the drawings, molded or cut-to-form, shall be used to provide shock and vibration protection to the contents of the case during transport. All items contained in the case shall be easily removed and replaced without damage to any item or to the cushioning material.

3.6.1.1 Set case finish. Unless otherwise specified (see 6.2), the case shall be finished in forest green in accordance with MIL-T-704. Metallic cases shall use the type D finishing system from table II of MIL-T-704 except that the final finish shall be in accordance with either MIL-E-52798 or MIL-E-52929. Glass fiber reinforced plastics or other compound non-metallic materials used for cases shall use the type C finishing system from table II of MIL-T-704 except that the final finish shall be in accordance with either MIL-E-52798 or MIL-E-52929. Non-metallic cases having the color pigmentation compounded in the material shall not require painting. A sample of either the forest green paint or of the case material having the color pigmentation compounded on it shall be submitted to the following laboratory for approval prior to acceptance of the initial production set; US Army Mobility Equipment Research and Development Command, Attn: DRDME-VO, Fort Belvoir, VA 22060. A minimum of a quart sample of paint from each production lot used must be forwarded. A sample of case material having the color pigmentation compounded in it shall be at least 2 inches square.

3.6.1.2 Catch assemblies. Unless otherwise specified (see 6.2), the catch assemblies per drawing 13222E7891 after treatment shall be painted at the option of the contractor.

3.6.2 Test-kit case. The components of the test kits for acidity, chloride, dissolved oxygen, iron and zinc shall be contained in impact-resistant plastic cases that are capable of being hand carried. Each case shall contain all the equipment required for conducting the test plus enough chemical reagents to conduct not less than 20 tests. Additional quantities of chemical reagents required to complete 100 tests shall be stored in the set case except for the fluoride test and the dissolved oxygen test. The kit cases shall be equipped with a molded or cut-to-form cushioning insert to provide shock and vibration protection and to permit easy removal and replacement of the components without damage to the component or the cushioning material. The cases shall be provided with latches that will withstand repeated use and will not be prone to accidental opening.

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3.7 Environmental.

3.7.1 Operating climatic conditions. The sets shall be capable of operating under the climatic conditions specified in table I when tested in accordance with 4.7.2.2.1.

TABLE I. Operating Climatic Conditions.

Climatic Category	Ambient Air Temp ° F	Solar Radiation Btu/ft ^{2} /hr	Ambient Relative Humidity, Percent
1	Nearly constant 75	Negligible	95 to 100
2	78 to 95	0 to 360	74 to 100
5	70 to 110	0 to 360	20 to 85

3.7.2 Storage climatic conditions. The sets shall be capable of storage under the climatic conditions specified in table II when tested in accordance with 4.7.2.2.2.

TABLE II. Storage Climatic Conditions.

Climatic Category	Induced Air Temp ° F	Induced Relative Humidity, Percent
1	Nearly constant 80	95 to 100
2	90 to 125° F	10 to 85
5	70 to 145° F	5 to 50
6	-10 to -30° F	Tending toward saturation

3.8 Marking.

3.8.1 Set cases. Each set shall be identified in accordance with MIL-STD-130. The identification marking shall be applied to the set case as specified on the drawings. The plates shall be attached to the case so that there will be no ingress of moisture and located so as to be both visible and legible.

3.8.1.1 Stenciling. Case tops will be stenciled with standard white stencil ink with the following "Storage Temp. Maximum 125° F". Stenciling will be applied on the case top siding above the case pressure relief valve with 1 inch high block letters.

3.8.2 Test-kit cases. Each test-kit case shall be identified in accordance with MIL-STD-130. The identification marking shall be applied to the case as specified herein and on the drawings. The identification shall be visible and legible.

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3.8.3 Reagent chemicals. Each container of reagent chemicals shall be labeled with the generic name or manufacturer's trade name and date of manufacture. Any reagent chemicals contained in the set such as sodium hydroxide solution, sodium thiosulfate reagent, zinc reagent or any other reagents which according to the manufacturer have a limited shelf life shall have as part of the label the expiration date of the reagent.

3.8.3.1 Cautions and warnings. Each container of reagent chemicals shall be labeled with cautions and warnings where contact with, inhalation of, or ingestion of the reagent would have an adverse effect on the user. Such information shall include but not be limited to the following:

(a) Generic name of the reagent.
(b) Symptoms.
(c) Antidote.
(d) Signal words (i.e., danger, warning, caution etc.).
(e) Statement of hazard (i.e., flammable, harmful, etc.).

Labeling of chemicals shall be in accordance with the requirements given in American National Standards Institute Bulletin ANSI Z129.1-1976, Precautionary Labeling of Hazardous Industrial Chemicals.

3.9 Instructions.

3.9.1 Loading diagram. The set case shall be provided with a loading diagram depicting the proper location and quantities of each item in the set case. The diagram shall be as specified on the drawing 13222E7917.

3.9.2 Instructions for test kits. Each kit shall be provided with instructions that contain detailed procedures for conducting the test. These instructions shall be printed on material which is water resistant or material which has been water resistant coated. Markings shall be in accordance with MIL-STD-130 and as specified on the drawings. These instructions shall be affixed to the interior lid of the kit with an adhesive conforming to MMM-A-1617. When this is not practical, the instructions shall be printed on material conforming to L-P-535, composition B, type I, class 3, matte finish, 0.020 inch thick (minimum), black printing on opaque white, on a sheet of material 8 inches wide x 10-1/2 inches long.

3.10 Transportability. The set shall be capable of withstanding the vibrational stress and shock encountered in transporting as loose cargo and airdrop delivery.

3.11 Weight and configuration. The weight of any case with contents shall not exceed 65 pounds. Cases shall be configured so that they can be easily carried by one man.

3.12 Safety. The components of the system must not present any hazard to personnel during production or use per MIL-STD-882.

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3.12.1 Safety Equipment. A safety pipette filler shall be provided with the set for use with the pipettes in transferring or measuring liquid reagents.

3.13 Human factors. The set shall conform to the human factors engineering requirements of MIL-STD-1472 and 3.4.3, 3.6, 3.8, 3.9, and 3.11 of this specification.

3.14 Workmanship. The cases and all component parts shall be fabricated free from any defects which could affect reliability, strength, or serviceability. The cushioning material installed in the cases shall be uniformly molded, formed, or cut. All markings, labels, and printed matter shall be legible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsbility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.1.2 End item final inspection report (EIFIR). Unless otherwise specified (see 6.2), each water testing kit shall be inspected in accordance with the applicable EIFIR (EF 13222E7900). One copy of the completed EIFIR shall be shipped with each inspected water quality analysis set.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Initial production inspection (see 4.4).
(c) Quality conformance inspection (see 4.5).
(d) Inspection comparison (see 4.6).
(e) Inspection of packaging (see 4.8).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction set shall be examined for the defects marked "X" in column 1 of table III. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The preproduction set shall be subjected to the tests marked "X" in column 1 of table IV. Failure of any test shall be cause for rejection.

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4.4 Initial production inspection. When specified (see 3.3), one or more initial production sets will be selected at random by the Government from the sets being produced by production tooling and will be examined for defects marked "X" in column 2 of table III and subjected to the tests marked "X" in column 2 of table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production set shall not exclude the remaining sets from the quality conformance inspection and acceptance provisions specified in Section 4. In addition to any test specified as part of the initial production test, the Government reserves the right to conduct any or all other tests contained in this specification as part of the initial production test, and failure of such additional tests shall have the same effect as failure of those tests specified as initial production tests.

4.4.1 Inspection failure. Failure of an initial production set to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the initial production set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the initial production inspection will be considered prima facie evidence that all sets accepted prior to the completion of the initial production inspection are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects marked "X" in column 3 of table III. AQL shall be 2.5 percent defective for major defects and 6.5 percent defective for minor defects.

4.5.3 Tests. Samples selected in accordance with 4.5.1 shall be subjected to the tests marked "X" in column 3 of table IV. AQL shall be 2.5 percent defective.

4.6 Inspection comparison. The Government may select sets at any time during the contract production period and subject these sets to the examinations marked "X" in column 4 of table III and the tests marked "X" in Column 4 of Table IV to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Governmenment, on sets selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production sets. In addition to any test specified as part of the inspection

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comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison set to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison set and shall be cause for refusal by the Government to continue acceptance of production sets until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on sets previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all sets accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection schedule.

4.7.1 Examination.

TABLE III. Examination Schedule.

Preproduction 1	Initial Production 2	Quality Conformance 3	Comparison 4	Defect 5		Requirement Paragraph 6
				Major		
X	X	X	X	101.	Clasps not as specified.	3.6.1
X	X	X	X	102.	Cushioning insert malformed or not cut to form.	3.6.1 and 3.6.2
X	X	X	X	103.	Latches not as specified.	3.6.2
X	X	X	X	104.	Components missing or not as specified.	3.5
X	X	X	X	105.	Materials not as specified.	3.4
X	X	X	X	106.	Components broken or damaged affecting serviceability.	3.4.1
X	X	X	X	107.	Materials showing evidence of degradation.	3.4.2

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TABLE III. Examination Schedule. (Cont'd)

Preproduction 1	Initial Production 2	Quality Conformance 3	Comparison 4		Defect 5	Requirement Paragraph 6
X	X	X	X	108.	Cautions, warnings, or antidote information missing, illegible, or incomplete.	3.8.3.1
X	X	X	X	109.	Procedural instructions missing or inadequate.	3.9
X	X	X	X	110.	Weight limits exceeded.	3.11
X	X	X	X	111.	Improper configuration for one-man carry.	3.11
X	X	X	X	112.	Workmanship not as specified.	3.14
X	X	X	X	113.	Expiration date of reagent exceeded.	3.8.3
X	X	X	X	114.	Case not finished as specified.	3.6.1.1
X	X	X	X	115.	Labels missing, not as specified, incomplete information or not legible.	3.8.3
				Minor		
X	X	X	X	201.	Reagent chemical quantities not as specified.	3.4.3
X	X	X	X	202.	Identification plates missing, not as specified, not attached as specified, obscured location, or not legible.	3.8
X	X	X	X	203.	Loading diagram not as specified, incomplete, or not legible.	3.9.1

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4.7.2 Tests.

4.7.2.1 Test schedule.

TABLE IV. Test schedule.

Preproduction 1	Initial Production 2	Quality Conformance 3	Comparison 4	Tests 5	Test Paragraph 6	Requirement Paragraph 7
X	X	X	X	Environmental	4.7.2.2	3.7
X	X	X	X	Transportability	4.7.2.3	3.10
X	X	X	X	Operational (precision)	4.7.2.4	3.5.1 thru 3.5.6
X	X	X	X	Safety	4.7.2.5	3.12
X	X	X	X	Human factors	4.7.2.6	3.13

4.7.2.2 Environmental.

4.7.2.2.1 Operation temperature. A certificate stating that test kits will be operational in the climatic conditions stated in table I will be acceptable in lieu of testing.

4.7.2.2.2 Storage. Each sample set shall be tested in accordance with MIL-STD-810, method 501, procedure II and method 502, procedure I, with the following exceptions:

Method 501, Procedure II

Step 4 - Change 68° C (154° F) to 52° C (125° F).
Step 6 - Repeat Steps 3, 4, and 5 one additional time (making a total of two 12 hour cycles).
Step 7 - Delete.
Step 8 - Delete.

Method 502, Procedure I

Step 2 - Lower the internal chamber temperature to -35° C (-30° F) for six continuous hours with no wind or solar radiation and with humidity tending towards saturation.
Step 3 - Delete.
Step 4 - Raise the internal chamber temperature to -23° C (-10° F) and maintain for 6 hours.
Step 5 - Repeat Steps 2 and 4 one additional time (making a total of two 12 hour cycles).

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The test sets will be transferred directly from the conditions of step 8, procedure I, method 501 to Step 1, procedure I, method 502. Upon completion of this test, each sample set will be inspected for damage to case and contents. Any evidence of permanent deformation, cracks, or broken items shall constitute failure of this test.

4.7.2.3 Transportability.

4.7.2.3.1 Loose cargo. Each sample set shall be tested in accordance with MIL-STD-810, category g, method 514.1, procedure XI (loose cargo) and method 516.1, procedure II (transient drop test). Upon completion of this test, each sample set will be examined for damage to the case and contents. Any evidence of permanent deformation, cracks, or broken items shall constitute failure of this test.

4.7.2.3.2 Airdrop. When specified (see 6.2), each sample shall be tested for airdrop capability and shall be prepared for test by strapping the test set with all components in place to a wood platform (approximately 30 inches by 30 inches) with energy dissipator material (MIL-H-9884, class 3) between the sample set and the wood platform. The total surface area of the dissipator material in contact with the test set shall be 64 square inches consisting of 8 pieces of dissipator material 4 inches x 4 inches x 3 inches high. The Dissipator material shall be arranged in four stacks of material each two pieces high. The stacks shall be glued to each other and to the case on the bottom edges of the case midway between the corners (fig. 1). The strapped case with all components, dissipator material, and wood platform shall be free dropped from a height of 12 feet 8 inches (152 inches) onto a flat rigid concrete surface. The drop shall be oriented so that the platform with test assembly shall fall flat at impact. Upon test completion, the set shall be examined for damage to case and contents. Any evidence of permanent deformation, cracks, or broken items shall constitute failure of this test.

4.7.2.4 Operational (accuracy). When specified (see 6.2), the following kits shall be removed from the sample sets and tested for the requirements as stated in the applicable paragraph as shown below. Unless otherwise specified (see 6.2), the contractor shall request from each vendor/supplier of chemicals, copies of their test data on each batch/distinct lot furnished to confirm that the chemicals meet specification requirements.

(a) Acidity test kit (see 3.5.1).
(b) High-range chloride test kit (see 3.5.2).
(c) Iron test kit (see 3.5.3).
(d) Dissolved oxygen test kit (see 3.5.4).
(e) Zinc test kit (see 3.5.5).
(f) Multipurpose test kit (see 3.5.6).

Test solutions will be prepared at each of the decision concentrations associated with each individual kit (i.e., 10, 20 and 150 mg/l in the case of sulfates). Replicate measurements will be conducted and the resulting precision will be calculated. Failure to meet applicable precision criteria shall constitute failure of this test.

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4.7.2.5 Safety. The set shall be evaluated throughout all testing to determine conformance to 3.12. Nonconformance to 3.12 shall constitute failure of this test.

4.7.2.6 Human factors. The set shall be evaluated throughout all testing to determine conformance to 3.13. Nonconformance to 3.13 shall constitute failure of this test.

4.8 Inspection of packaging.

4.8.1 Quality conformance inspection of pack.

4.8.1.1 Unit of product. For purpose of inspection, a unit of product shall be a completed pack of a single set prepared for shipment.

4.8.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.1.3 Examination. Samples selected in accordance with 4.8.1.2 shall be examined for the following major defects. AQL shall be 2.5 percent defective. All completed pallet loads shall be examined for the applicable defects.

116. Set components not fitted securely in designated spaces within carrying cases.
117. Carrying case clasp not properly latched.
118. Shipping container not as specifed for levels A and B.
119. Shipping container not strapped as specified for levels A and B.
120. Palletized loads not on type of pallet specified for level A or B or commercial.
121. Sets not secured to pallets as specified for all degrees.
122. Marking illegible, incomplete, incorrect or missing.

5. PACKAGING

5.1 Preservation level A, level B, or commercial. The carrying case shall be checked to insure that all set components are securely fitted into their designated locations. All clasps shall be securely latched.

5.2 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

5.2.1 Level A. The carrying cases comprising a complete set shall be packed in a snug-fitting box conforming to PPP-B-601, overseas type, style I. The box shall be closed and strapped in accordance with the appendix to the applicable box specification, except that the strapping shall conform to QQ-S-781, class 1, type I or IV, minimum size not less than 1/2 inch wide x 0.02 inch thick; and, unless otherwise specified herein, shall be finish B. When specified (see 6.2), strapping shall be finish A.

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5.2.2 Level B. The carrying case comprising a complete set shall be packed as specified in 5.2.1, except that the box shall conform to PPP-B-636, type CF, class weather resistant, variety DW, grade V11c. The box shall be closed, sealed, and reinforced in accordance with the appropriate requirements of the box specification. Strapping shall be flat-steel type.

5.2.3 Commercial. The carrying case comprising a complete set shall be packed together in accordance with MIL-STD-1188.

5.3 Palletization. When four or more sets are to be shipped to the same destination, packed carrying cases shall be palletized. For levels A and B, pallets shall conform to NN-P-71, type IV. Loading and securing of carrying cases shall be in accordance with MIL-STD-147. For commercial, pallets shall be a commercial expendable wood type, 40 inches by 48 inches in size. Flat steel strapping shall be used to secure carrying cases to pallet.

5.4 Marking. Marking for level A and level B shall be in accordance with MIL-STD-129. Commercial marking shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The Water Quality Analysis Set, Preventive Medicine, is intended for use by medical personnel for on-site field monitoring of the quality of raw water sources, wastewater effluents, and the drinking water produced to insure protection of personnel from waterborne diseases and effects of toxic chemicals.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
(c) Time frame required for submission of preproduction model, and number of sets required (see 3.2).
(d) When the Government will conduct any or all of the preproduction model examination and tests. When the Government will conduct some but not all of the preproduction examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
(e) When initial production inspection is required and number of sets to be furnished (see 3.3).
(f) When color/finish is other than as specified (see 3.6.1.1).
(g) When painting of catch assembly is required (see 3.6.1.2).
(h) When an EIFIR is not required for each set (see 4.1.2).
(i) When set is to be airdrop tested (see 4.7.2.3.2).
(j) When kits shall be removed and tested (see 4.7.2.4).
(k) When vendor data is not required (see 4.7.2.4).
(l) Degree of packing required (see 5.2).
(m) When strapping shall be finish A (see 5.2.1).

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6.3 Preproduction model. Any changes or deviations of production sets from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the contractor of his obligation to furnish sets conforming to this specification (see 3.2).

6.4 Supplies and equipment. The contracting officer should furnish the contractor the list of supplies and equipment as specified in SC 6630-97-CL-E08, NSN 6630-00-140-7826.

6.5 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, and contractor's operation manual to be furnished with each set.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.4).

Custodian:
Army - ME

Review activity:
Army - MD

User activity:
Army - CE

Preparing activity:
Army - ME

Project 6630-A326

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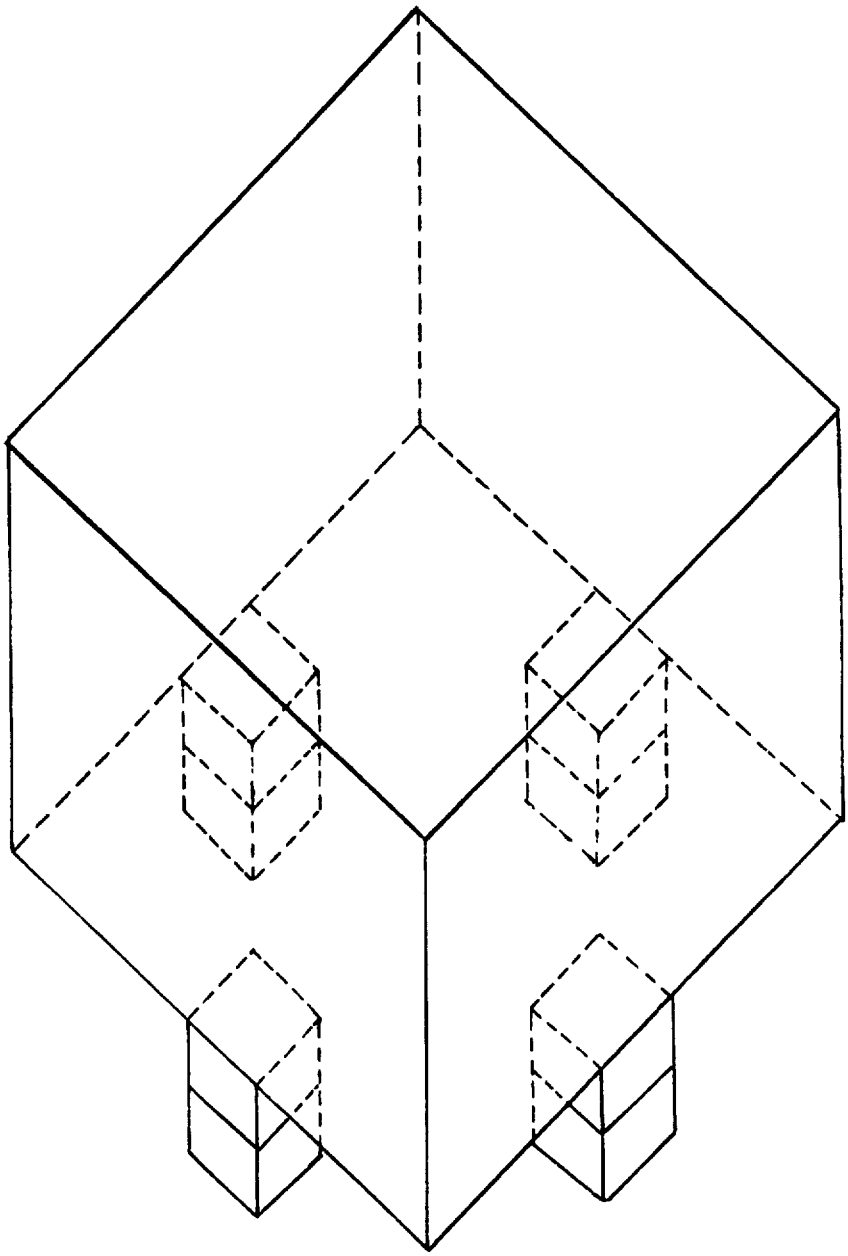


FIGURE 1. Energy dissipator material.

X-3527

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-W-52875B(ME) Water Quality Analysis Set: Preventive Medicine

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

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